

30-DAY HOSPITAL READMISSION

A domain-informed, leakage-safe prediction study

Baseline utilization • index-diagnosis EHR • encounter-linked SDOH

97,310 encounters | 59,449 patients | 11,973 readmissions

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MOTIVATION

Why ED & Inp. readmission and SDoH matter across Kansas and Sinai

Why readmission matters

CLINICAL BURDEN

- A 30-day return can signal unresolved illness, complications, or gaps in follow-up.
- Earlier risk identification can support discharge planning and targeted outreach.
- Preventing avoidable returns can improve continuity of care and reduce health-system burden.

Why SDOH Matters

- Housing, food, transportation, and financial barriers can prevent patients from following a care plan.
- SDoH add context that diagnoses and prior utilization alone may miss.
- They also identify needs that hospitals and Community Health Workers can address through referrals and support

Kansas and Sinai

- The analyzed encounter-level EHR cohort came from Stormont Vail Health; Kansas

97,310

>1,300

Kansas encounters

Sinai pilot patients

- Both studies combine clinical information with SDoH to estimate 30-day ED readmission risk.
- Kansas provides a large EHR cohort across Stormont Vail Health. Sinai adds direct Community Health Worker engagement in under-resourced Chicago communities.

MOTIVATION

Sinai showed that Community Health Worker and SDoH information can improve prediction. The larger Kansas cohort tests whether those insights extend to a different population and health system.

STUDY QUESTIONS

Two related—but distinct—analyses

Primary analysis

Can information available by index discharge predict qualifying hospital readmission within 30 days?

Baseline → Baseline + EHR → Baseline + EHR + SDOH

Secondary analysis

Among known 30-day readmissions, can index-available predictors distinguish a different-body-system return from a clinically related return?

This predicts return type—not readmission risk.

DATA SOURCE AND COHORT

One Stormont Vail Health cohort powers every comparison

97,310

ELIGIBLE INDEX ENCOUNTERS

59,449

UNIQUE PATIENTS

11,973

30-DAY READMISSIONS

Source and linkage

Stormont Vail Health patient-level EHR, January 2022–December 2025.

EncounterKey + PatientDurableKey link index encounters; PrimaryDiagnosisKey → DiagnosisKey resolves ICD-10-CM; DepartmentKey identifies facility; SDOH links at patient and encounter level.

All cohort EncounterKeys matched original encounter cells (100% linkage coverage).

Clinical meaning guided feature construction

Domain-informed decisions

- Prespecified prior-utilization indicator.
- ICD-10-CM summarized through AHRQ CCSR.
- Literature-informed alcohol, activity, stress, and resource-risk cutoffs.
- Explicit unable, nonresponse, no-record, and conflict states.

Why not purely data-driven?

- Clinicians and community health workers need interpretable definitions.
- Prespecification limits recording after seeing outcomes.
- Transparent categories expose missingness and documentation patterns.
- Model selection still used validation performance.

Data preprocessing: 2023 paper vs. current project

Pipeline replaces inferred values and changing cohorts with explicit states, fixed encounters, and leakage-safe fitting.

Preprocessing decision	2023 Karam et al. paper	Current project
Missing SDOH responses	Mode-imputed among engaged patients; refusal treated as NA; some “unsure” responses recoded as risk.	Preserves risk, negative, unable, unresolved, no-record, and eligible conflict/partial states separately.
Feature construction	Aggregates 10 contact logs; creates engagement duration, total time, contact totals, and repeat-referral count.	Uses prior 365-day acute-care utilization, CCSR body systems, and reduced mutually exclusive SDOH domains.
Sparse and redundant data	Drops mostly empty SDOH variables and correlation-based redundancies; manually collapses race and insurance.	Uses development-only frequency/event thresholds; removes exact aliases and development-constant features.
Timing / leakage control	Cumulative CHW contacts and case duration may extend into the 30-day outcome window; timing restriction is not stated.	Only pre-index or index-time predictors; excludes post-index responses, next-admission fields, and outcome-derived data.
Split and transformation fitting	80/20 split with 10-fold CV; patient-level separation and fold-specific imputation/scaling are not reported.	Locked patient-level train/validation/test split; median imputation and all model choices fit without test data.

WHAT CHANGED

Missingness stays informative

No response is not automatically treated as a negative answer.

Comparisons use one cohort

M0, M1, and M2 retain the same 97,310 encounters.

Validation is patient-safe

The same patient cannot appear in both development and test data.

Sources: Karam et al., TransAI 2023; Current M0/M1/M2 protocol and reproducible methods log (Aug. 2026)

EDA- domain/ literature informed preprocessing stratifies all 12 SDOH

Exact patient–encounter linkage preserved adverse, reference, nonresponse, conflict, and no-record states.

ALCOHOL High • low • none High/lower use → risk; none → reference. CDC excessive drinking • LOINC AUDIT-C	IPV Positive • negative Any Yes across four questions overrides No/nonresponse. CMS AHC HRSN Tool (2021)
DEPRESSION Positive • negative PHQ-2 ≥3 or EPDS ≥8 → positive. Kroenke et al. (2003) • Matijasevich et al. (2014)	SMOKING Current • former • never planned EDA: exact matches were unknown; retained nonresponse/no record. Project encounter-level EDA • LOINC SDOH (2018)
FINANCIAL STRAIN Positive • negative Rent/mortgage difficulty or shutoff evidence → risk. CMS AHC HRSN Tool (2021)	UTILITIES HARSHIP Positive • negative Somewhat hard+ paying for basic needs → risk. Tucher et al. (2024)
FOOD INSECURITY Positive • negative Either Hunger Vital Sign item adverse → positive. Hager et al. (2010)	SOCIAL CONNECTION None • moderate • frequent Frequent contact → reference; partner status retained. Fujiwara et al. (2022) • Tomioka et al. (2018)
HOUSING Positive • negative Homeless/unstable housing or multiple moves → risk. CMS AHC HRSN Tool (2021)	TRANSPORTATION Positive • negative Transport prevented care, daily needs, or work → positive. CMS AHC HRSN Tool (2021)
PHYSICAL ACTIVITY 0 • 1–149 • ≥150 min/wk Exercise Vital Sign = days/week × minutes/day. Coleman et al. (2012) • Piercy et al. (2018)	STRESS None/low • moderate • high Moderate starts at “to some extent”; high = rather/very much. Elo et al. (2003) • Wiegner et al. (2015)
EDA + PREPROCESSING CHOICE Each domain became one mutually exclusive state: risk • clean negative • unable • nonresponse • no record • conflict/partial. Unknown was never coded negative. Sparse conflict/partial remained separate only when development n ≥100 and readmissions ≥20. Project M2 protocol • CMS AHC HRSN Tool (2021) • Regenstrief LOINC SDOH (2018)	

Profiling established grain, linkage, and usable states

Encounter CSV

7,675,801 rows • 31 columns

EncounterKey
PatientDurableKey
Date
PrimaryDiagnosisKey
Type / VisitTypeDescription
DepartmentKey
IsEdVisit
IsHospitalAdmission
IsInpatientAdmission
IsObservation

Diagnosis CSV

1,531,262 rows

DiagnosisKey
DiagnosisValue
GroupCode / GroupName
DiagnosisName

1,382,556 unique keys
129,173 reused/ambiguous keys
99,261 distinct ICD-10 values

SDOH CSV

3,977,901 response rows

DisplayName • AnswerText • Domain
EncounterKey • PatientDurableKey

32 question labels
125 raw answers
Examples: Yes, No, Never, Former,
Current, declined, unable, unknown

Three matrices isolate incremental information

M0 • Baseline

1 predictor

Prior_High_Combined_Utilizer_365d

Intentional comparator: ≥ 4 distinct prior acute-care days in the preceding 365 days.

M1 • Baseline + EHR

23 logistic • 24 tree

M0 + index diagnosis body-system indicators + diagnosis-quality states.

Logistic omits circulatory as reference.

M2 • Full model

75 logistic • 87 tree

M1 + 12 SDOH domains encoded as risk, reference, unable, nonresponse, no-record, or conflict states.

The denominator and patient split were fixed once

62,886

TRAIN

14,622

VALIDATION

19,802

HELD-OUT TEST

Controls applied to every model

- Patient overlap across partitions = 0.
- Only pre-index or index-discharge information was eligible.
- No next-admission diagnosis, post-index response, or test-derived information entered prediction.
- Preprocessing was fitted inside training; validation selected settings and thresholds; test opened once.

What the reported metrics mean

AUROC

Ranking ability: how often an event receives a higher score than a non-event.

0.50 = chance; higher is better.

Average precision + recall

Precision asks how many flagged encounters truly had the outcome.

Recall asks how many actual outcomes were detected.

Interpret under 12.1% prevalence.

Brier + calibration

Brier score measures probability error; lower is better.

Calibration checks whether predicted probabilities agree with observed risk.

PRIMARY RESULTS

EHR supplied the major gain; SDOH added a smaller increment

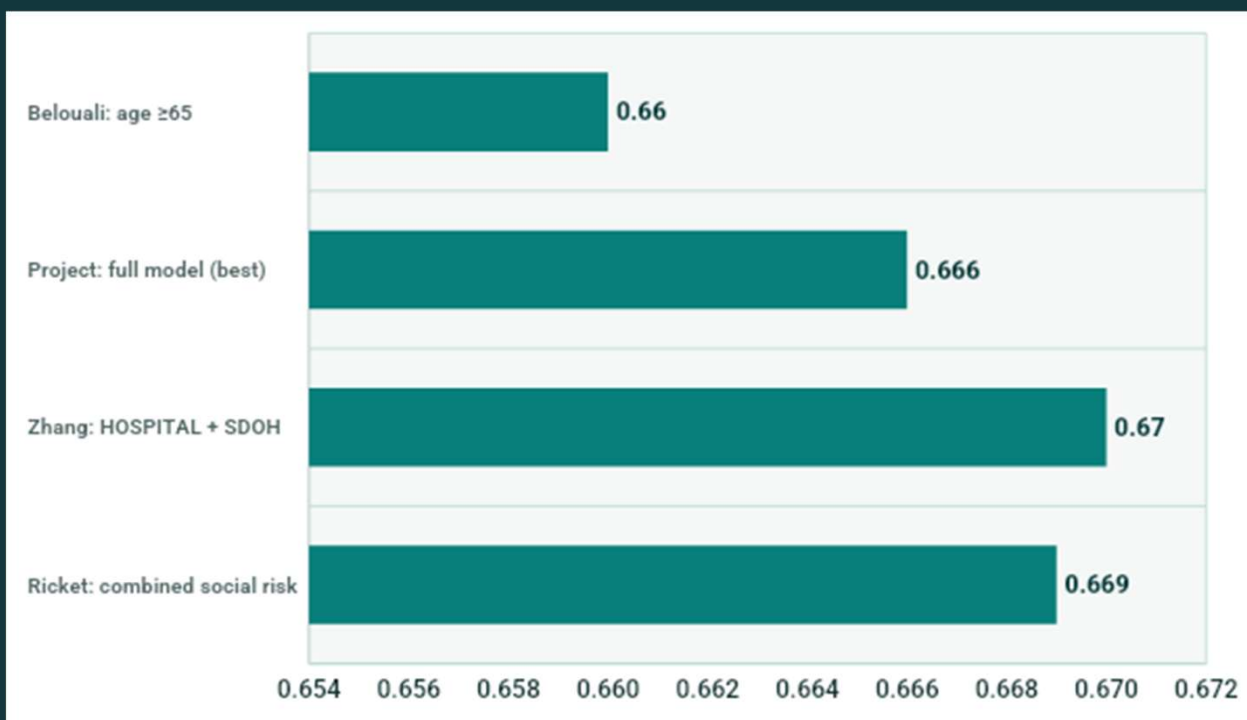
Held-out test n = 19,802 • ranges across four algorithms • recall/precision at validation-selected Youden

Stage	AUROC (95% CI envelope)	Recall	Precision
M0 • Baseline	0.569 (0.556–0.584)	22.4%	26.6%
M1 • Baseline + EHR	0.648–0.651 (0.633–0.666)	40.6–43.0%	22.8–23.5%
M2 • Full model (Baseline + EHR + SDOH)	0.661–0.666 (0.644–0.681)	44.8–51.5%	21.0–23.7%

EHR supplied the major discrimination gain (about +0.08 AUROC). SDOH added a smaller increment. CI envelopes summarize the lowest lower bound and highest upper bound across the four model-specific patient-bootstrap intervals.

DISCRIMINATION

Held-out AUROC sits inside the closest published range



0.661–0.666

project AUROC range

Ricket et al. reported 0.635–0.672 across clinical, social-risk, and combined models. The closest combined score was 0.669.

Comparable—not identical populations or outcomes.

Index-diagnosis EHR features added substantial discrimination

Patient-separated held-out test n = 19,802 • M0 = prior utilization • M1 = M0 + index-diagnosis EHR

Model	M0 AUROC	EHR-only AUROC	M1 AUROC (95% CI)	M1 sensitivity (95% CI)	M1 avg. precision	M1 Brier
Logistic regression	0.569	0.610	0.648 (0.633–0.663)	40.6% (37.8–44.0)	0.230	0.101
HistGradientBoosting	0.569	0.609	0.651 (0.636–0.665)	43.0% (40.1–46.1)	0.230	0.101
Random forest	0.569	0.609	0.651 (0.635–0.665)	40.6% (37.9–43.5)	0.230	0.101
XGBoost	0.569	0.609	0.651 (0.634–0.666)	43.0% (39.7–45.8)	0.231	0.101

EHR increment

EHR improved AUROC by about 0.08 over M0 across every classifier family.

Sensitivity and uncertainty

Sensitivity is reported at the validation-selected Youden threshold.
95% CIs use 200 patient-level bootstrap replicates.

The EHR block improved ranking performance; differences among M1 model families remained minimal.

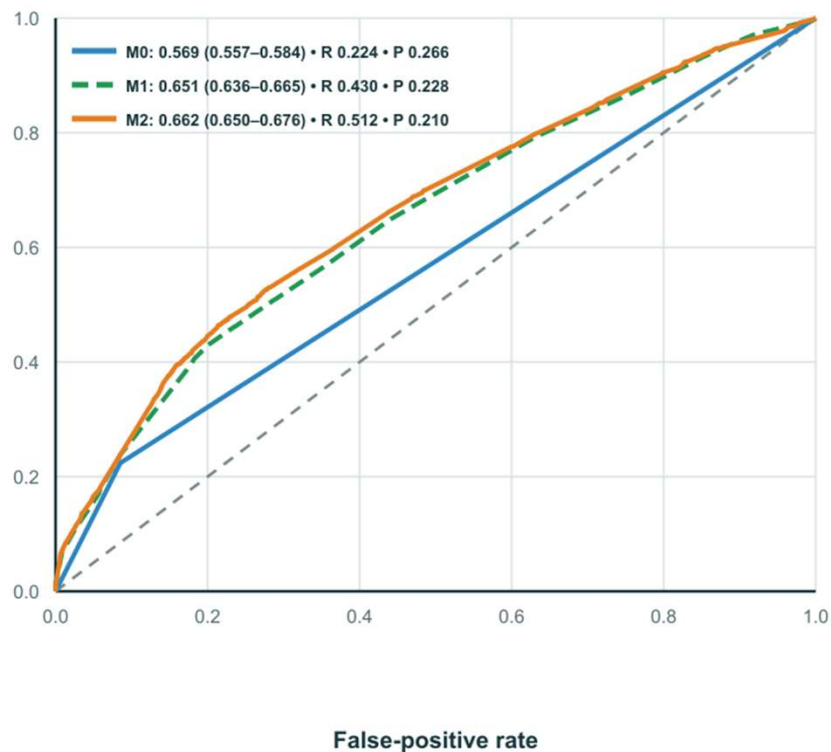
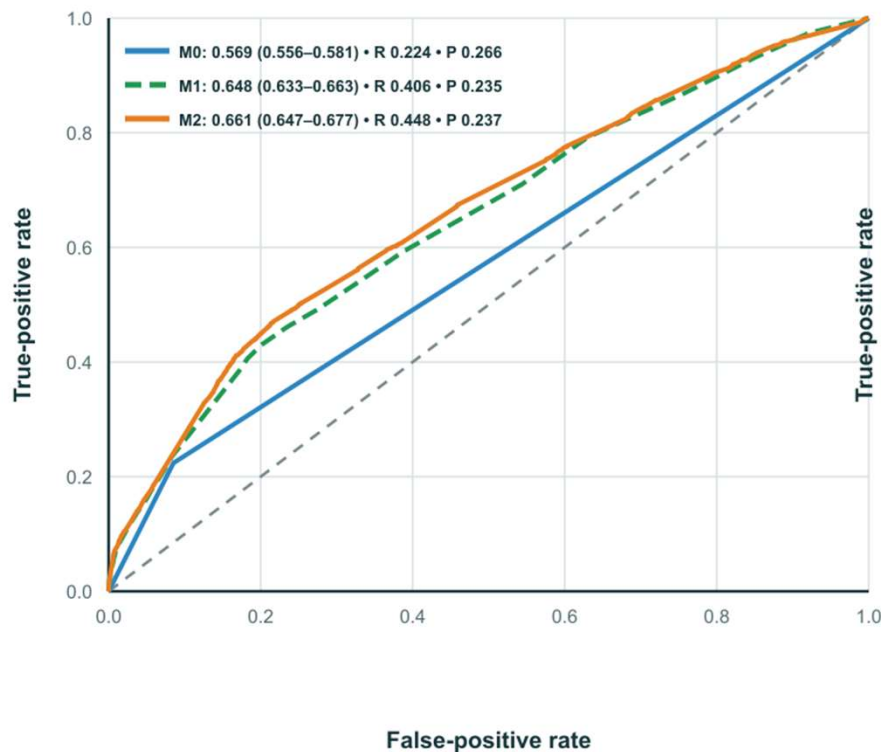
- AUROC = discrimination across all thresholds • Sensitivity = share of readmissions detected at the selected threshold
- Average precision summarizes precision–recall performance
 - Lower Brier scores indicate better probability accuracy

Incremental ROC: logistic and histogram boosting

Held-out test: 19,802 encounters • 2,404 readmissions • identical patient-separated cohort
AUROC (95% CI) • recall (R) and precision (P) at validation-selected Youden threshold

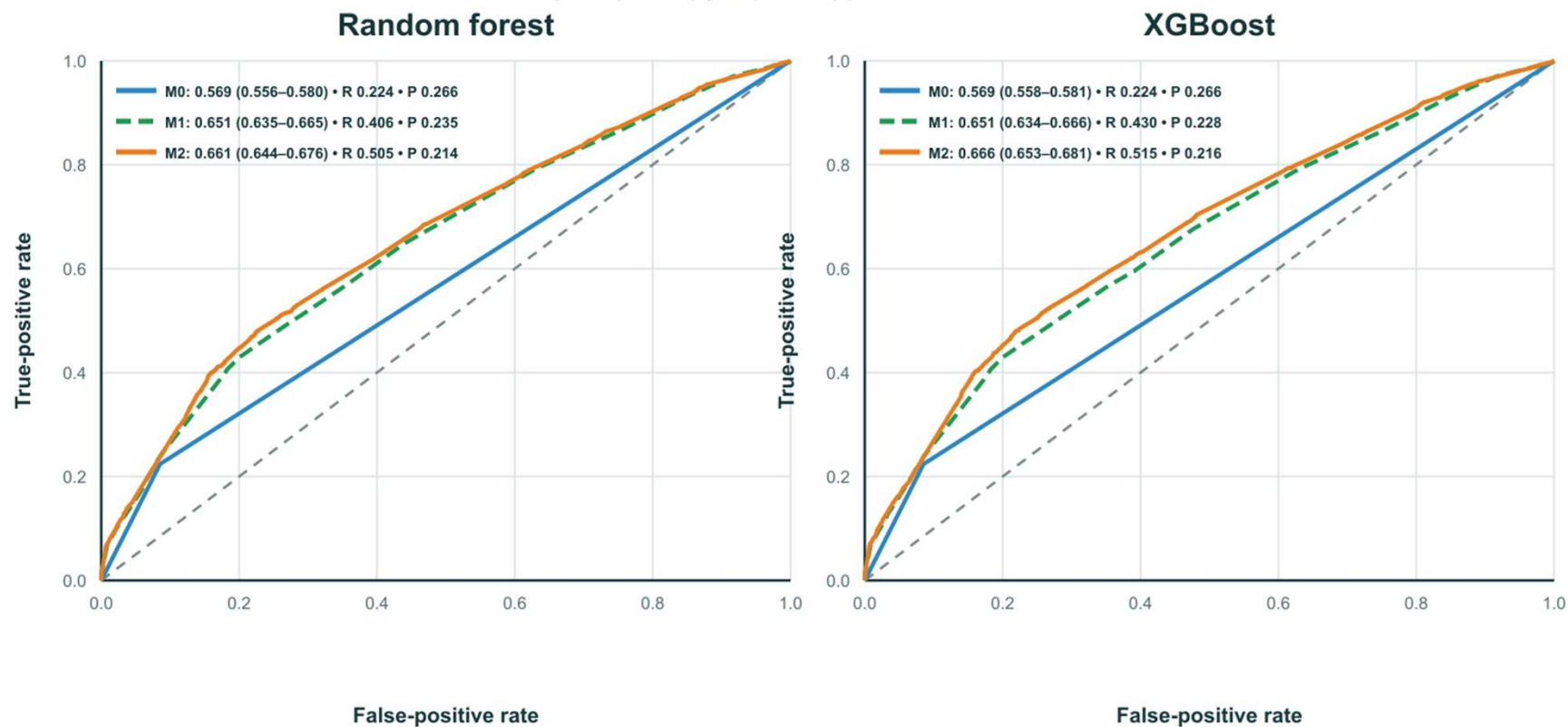
Logistic regression

HistGradientBoosting



Incremental ROC: random forest and XGBoost

Held-out test: 19,802 encounters • 2,404 readmissions • identical patient-separated cohort
AUROC (95% CI) • recall (R) and precision (P) at validation-selected Youden threshold



FEATURE IMPORTANCE

Utilization and diagnosis context dominate

Mean held-out permutation importance across four model families



Δ AUROC is a disruption measure—not an individual feature AUROC.

Prior utilization and diagnosis systems dominate the consensus top 20

Mean held-out permutation importance across logistic regression, HistGradientBoosting, random forest, and XGBoost

Rank	Feature	Feature family / CCSR body system	Mean	Rank	Feature	Feature family / CCSR body system	Mean
Delta AUROC				Delta AUROC			
1	Prior high combined utilizer (365 d)	Utilization	0.050 0	11	Body system unclassified	CCSR body system: Unclassified	0.003 5
2	Pregnancy-related diagnosis	CCSR body system: Pregnancy	0.041 2	12	Smoking - nonresponse	SDOH documentation	0.003 2
3	Food insecurity - no record	SDOH documentation	0.016 5	13	Mental/behavioral diagnosis	CCSR body system: Mental/behavioral	0.002 9
4	Perinatal diagnosis	CCSR body system: Perinatal	0.013 4	14	Financial strain - no record	SDOH documentation	0.002 8
5	Physical activity risk	SDOH measured risk	0.007 1	15	Smoking - no record	SDOH documentation	0.002 4
6	Diagnosis missing	Diagnosis quality	0.006 4	16	Factors influencing health status	CCSR body system: Factors/contacts	0.002 1
7	Depression - no record	SDOH documentation	0.006 0	17	Nervous system diagnosis	CCSR body system: Nervous system	0.001 8
8	IPV - no record	SDOH documentation	0.004 7	18	Neoplasm diagnosis	CCSR body system: Neoplasms	0.001 7
9	Transportation - no record	SDOH documentation	0.004 4	19	Alcohol - no record	SDOH documentation	0.001 7
10	Utilities - no record	SDOH documentation	0.004 2	20	Stress - no record	SDOH documentation	0.001 0

CCSR note: ccsr_system_* predictors are broad body-system indicators, not detailed CCSR diagnosis categories. Delta AUROC is a disruption measure, not an individual feature AUROC.

Four algorithms performed similarly on held-out data

Logistic regression

AUROC 0.661 (0.647–0.677)

Recall 44.8% • Precision 23.7%

Average precision 0.245

Brier 0.101 • Youden 0.126

HistGradientBoosting

AUROC 0.662 (0.650–0.676)

Recall 51.2% • Precision 21.0%

Average precision 0.244

Brier 0.101 • Youden 0.110

Random forest

AUROC 0.661 (0.644–0.676)

Recall 50.5% • Precision 21.4%

Average precision 0.242

Brier 0.101 • Youden 0.118

XGBoost

AUROC **0.666** (0.653–0.681)

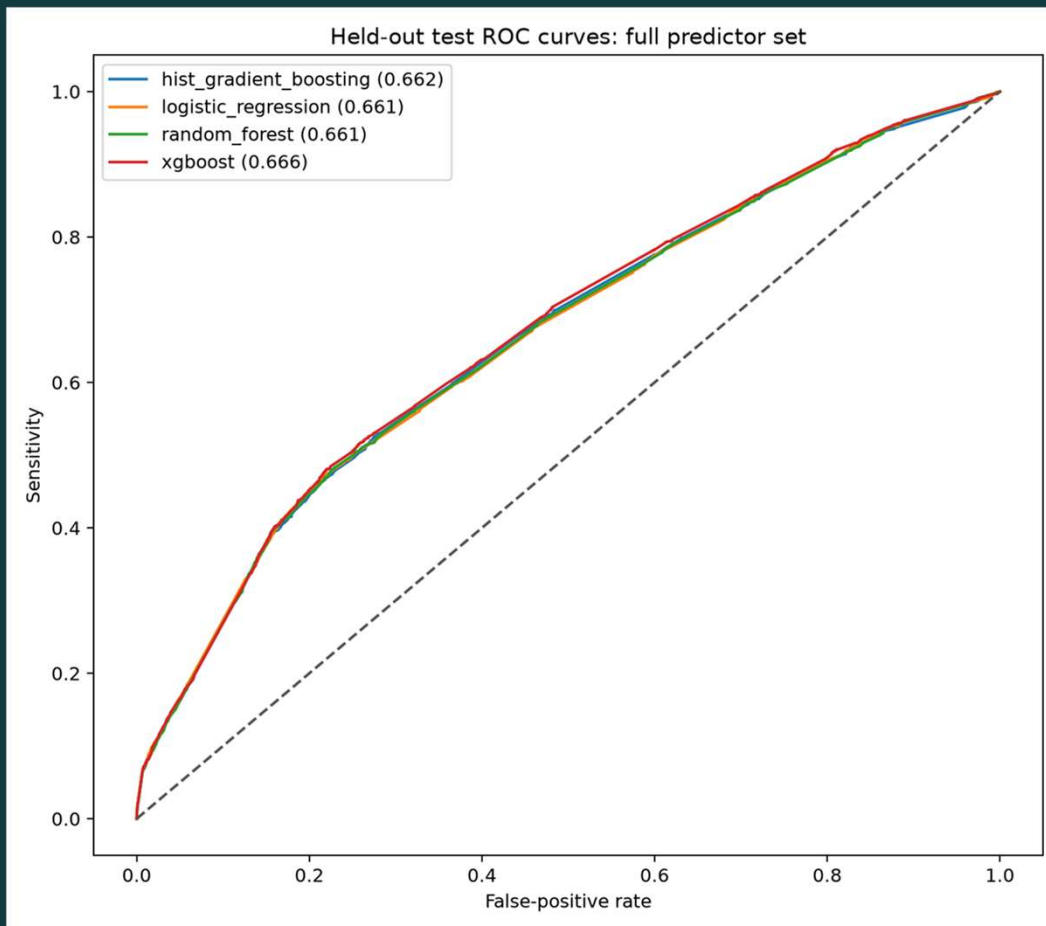
Recall 51.5% • Precision 21.6%

Average precision 0.246

Brier 0.100 • Youden 0.108

XGBoost ranked best, but model-family differences were narrow.

Held-out ROC curves — full M2 models



Full M2 • AUROC (95% CI) • recall • precision

Logistic 0.661 (0.647–0.677) • R 44.8% • P 23.7%

HistGB 0.662 (0.650–0.676) • R 51.2% • P 21.0%

Random forest 0.661 (0.644–0.676) • R 50.5% • P 21.4%

XGBoost 0.666 (0.653–0.681) • R 51.5% • P 21.6%

R/P use validation-selected Youden thresholds.

Test n=19,802; readmissions=2,404 (12.14%).

Source: heldout_test_predictions.csv.gz

95% CI: 200 patient bootstraps • R/P: validation-selected Youden

Sensitivity changed who was flagged—not how the model was trained

Hyperparameters were selected by validation AUROC; sensitivity was prioritized afterward by lowering the classification threshold.

TRAINING OBJECTIVE

Maximize validation AUROC

The models learned to rank patients by 30-day readmission risk across all possible thresholds.

Full XGBoost AUROC
0.666



POST-TRAINING THRESHOLD CHOICE

Lower cutoffs catch more readmissions—but create more false alerts

	SENSITIVITY	SPECIFICITY
Youden Balanced operating point	51.5%	74.2%
80% target Prioritize fewer missed cases	79.4%	38.4%
90% target Highest recall setting	86.3%	27.3%

Models were trained and tuned for AUROC. Sensitivity was prioritized afterward through validation-selected 80% and 90% operating thresholds to reduce missed readmissions.

SECONDARY ANALYSIS HYPOTHESIS

Can we predict a different body system versus clinically related return?

Outcome and population

Known 30-day readmissions only.

Positive: different body system.

Comparison: exact ICD-10, same CCSR category, or same body system.

Unknown relationships excluded from primary inference and prediction.

Leakage-safe workflow

Index-available M2 predictors only.

Original patient-separated partitions retained.

Validation AUROC selected settings; test opened once.

Time-to-return: Kruskal–Wallis, then Dunn tests with Holm correction.

SECONDARY ANALYSIS MOTIVATION

Literature supports All-cause readmission hides clinically different return pathways

1 The outcome is heterogeneous

10%–35%

Same-condition returns in three major Medicare cohorts

- After heart failure, acute myocardial infarction, or pneumonia, most 30-day readmissions involved another diagnosis. This supports analyzing return type rather than treating every return as one outcome.¹
- AHRQ likewise defines all-cause readmission without requiring the same diagnosis as the index stay.²

2 Relatedness has clinical meaning

- Published work classifies a return as related when index and readmission diagnoses involve the same organ system. This directly supports our body-system comparison.³
- In hemodialysis, the adjusted probability of an unrelated return rose from 19.1% to 31.2% as prior hospitalization burden increased.³
- The distinction separates recurrence of the index problem from broader vulnerability, multimorbidity, or another acute process.

3 The predictors match the hypothesis

- Prior utilization is relevant because hospitalization burden predicts unrelated returns.³
- Index diagnosis defines the clinical relationship between the two encounters.
- SDoH capture cross-cutting barriers. Transportation, affordability, food insecurity, and neighborhood disadvantage have been associated with 30-day readmission.^{4–6}

SECONDARY QUESTION

Among known 30-day readmissions, can information available at index discharge distinguish a different-body-system return from a clinically related return?

SECONDARY ANALYSIS COHORT

The secondary cohort includes both lower and high prior utilizers

11,973

readmitted index encounters
7,570 unique patients

- 407

unknown diagnosis relationship

=

11,566

analytic encounters
7,452 unique patients

High prior utilization was a predictor, not an eligibility rule: ≥4 distinct ED or inpatient acute-care dates in the 365 days before each index encounter.

Prior-utilization group	Encounters	Unique patients	Different body system	Clinically related
Lower: 0–3 prior dates	8,851	6,855	3,848 (43.5%)	5,003 (56.5%)
High: ≥4 prior dates	2,715	1,193	1,280 (47.1%)	1,435 (52.9%)
All known relationships	11,566	7,452	5,128 (44.3%)	6,438 (55.7%)

What the table shows

Most analytic encounters came from the lower-utilization group. High utilizers contributed 2,715 encounters, or 23.5% of the analytic cohort.

Different-body-system returns occurred in both groups and were somewhat more frequent among high-utilizer encounters (47.1% versus 43.5%).

The cohort was not restricted to high utilizers

Prior utilization entered the M2 model as a binary index-available feature.

Patient counts by utilization group are not additive: 596 patients contributed encounters in both groups at different times.
Clinically related combines **exact ICD-10, same CCSR category, and same body system**.

Return timing and diagnosis relationship differed

Time to readmission

Different system: n=5,128 • **median 14 days**
(suggesting possible comorbidity/ clinical complexity.)

Exact ICD-10: n=2,045 • median 13 days

Same CCSR: n=1,315 • median 12 days

Same system: n=3,078 • median 9 days

Kruskal–Wallis $H(3)=346.54$; $p<0.001$.

Five of six adjusted comparisons significant; exact ICD vs same CCSR was not ($p=0.052$).

Leakage-safe prediction

Held-out n=2,320 • positive prevalence 45.4%

AUROC (95% CI) • **recall** • **precision**

Random forest **0.810** (0.789–0.829) • 97.4% • 62.2%

HistGradientBoosting **0.810** (0.788–0.830) • 95.3% • 63.2%

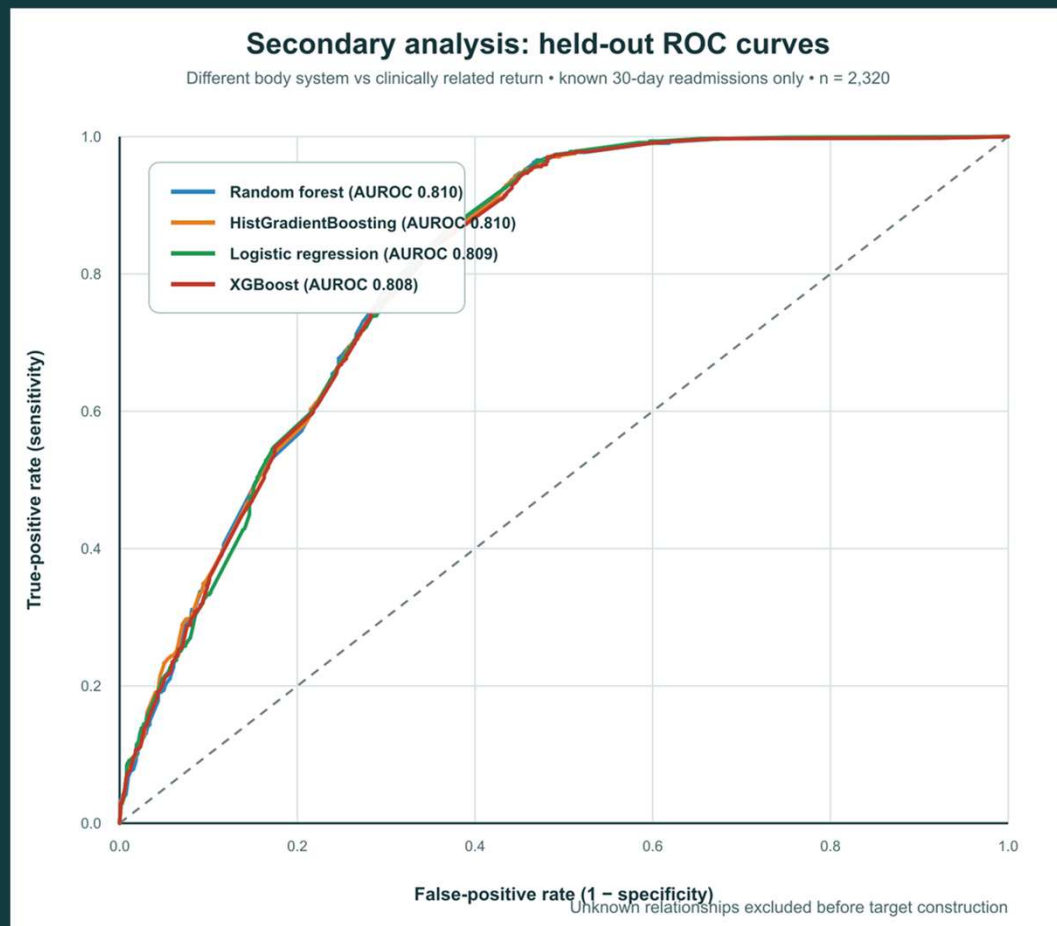
Logistic regression **0.809** (0.786–0.829) • 86.1% • 66.3%

XGBoost 0.**808** (0.786–0.827) • 95.9% • 62.6%

Recall/precision use validation-selected Youden thresholds.

No formal pairwise AUROC test.

ROC curves for diagnosis-system change



Different body system vs clinically related return

Test n=2,320 • positive=1,054 (45.4%)

Related=1,266: exact 407; same CCSR 265; same system 594

AUROC (95% CI) • recall • precision

Random forest **.810** (.789–.829) • 97.4% • 62.2%

HistGB .810 (.788–.830) • 95.3% • 63.2%

Logistic .809 (.786–.829) • 86.1% • 66.3%

XGBoost .808 (.786–.827) • 95.9% • 62.6%

Recall/precision at validation-selected Youden.

Unknown excluded; index-available M2 predictors only.

Source: diagnosis_relationship_test_predictions.csv.gz
AUROC uses 500 patient-bootstrap replicates; thresholds recovered from the original validation design

Decision log 1/3: cohort, time, and linkage

Row grain, follow-up, and temporal boundaries were fixed before modeling; project-specific operationalizations are labeled.

Decision	Choice used in this build	Evidence or status
Data source	Stormont Vail Health EHR, Jan 2022–Dec 2025	Project provenance audit
Analytic unit	One eligible index hospital encounter per PatientDurableKey + EncounterKey; same 97,310 rows in M0–M2.	Canonical cohort protocol; one-to-one grain checks
Time basis	Encounter Date/discharge fields—not instant timestamps—with complete 30-day follow-up.	Project source-schema and eligibility decision
Primary outcome	First subsequent qualifying hospital admission within 30 days; 60/90-day outcomes excluded.	CMS 30-day measure (2011); local encounter-level implementation
Next-admission link	Next_Hospital_Admission_EncounterKey → copied encounters, checked within patient; next diagnosis used only secondarily.	Project linkage QA; all next-admission fields excluded from predictors

Decision log 2/3: feature construction and encoding

Clinical meaning was preserved while ambiguous, missing, and sparse states remained auditable.

Decision	Choice used in this build	Evidence or status
Prior utilization	≥4 distinct ED or hospital-use dates in [index–365 days, index); same-day encounters counted once.	Health Quality Ontario (2022); combined-use extension is project-specific
Diagnosis linkage	PrimaryDiagnosisKey → DiagnosisKey → normalized ICD-10-CM; conflicting key maps flagged ambiguous.	Project diagnosis QA; no invented tie-breaker
Clinical grouping	ICD-10-CM → AHRQ CCSR default category → body system; unresolved/unclassified rows retained.	AHRQ CCSR v2026.1
SDOH construction	Left join on PatientDurableKey + EncounterKey; 12 domains encoded as mutually exclusive states.	Regenstrief LOINC (2018); CMS AHC HRSN (2021); Chen et al. (2020)
Feature eligibility	Conflict retained only if development $n \geq 100$ and events ≥ 20 ; exact aliases and development constants removed.	Project coefficient-stability rule; test data never consulted

What the analysis supports—and what it does not

Supported

- EHR diagnosis context substantially improves ranking.
- SDOH adds a smaller increment.
- Return reason and timing vary across diagnosis relationships.

Limitations

- Single health system.
- Moderate discrimination.
- Uneven screening availability.
- Secondary model applies only after readmission.

Next steps

- External and temporal validation.
- Subgroup calibration review.
- Workflow-specific threshold choice.
- Prospective evaluation of capacity and harms.

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SUMMER RESEARCH SUPPORT

DREAM made this summer research possible

- NSF DREAM—Distributed REsearch Apprenticeships for Master's—is administered by CRA-WP.
- It supports master's students entering computing from non-computing bachelor's backgrounds, with an emphasis on widening participation in tech.
- Participants complete a mentored 3–4 month research project with stipend support, professional development, and cohort activities.
- Its purpose is to build research confidence and widen pathways into computing research and potential PhD study.



DREAM

research
mentoring
community



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WITH GRATITUDE

Thank you for welcoming me into the work

- Thank you to the VIDA group at DePaul for welcoming me into the team and its research community this summer.
- Special thanks to Dr. Tchoua for graciously guiding my work and facilitating such a meaningful research experience.
- I gained hands-on learning in computer vision, machine learning, and data-science methods—from data preparation through model evaluation and communication—and learned how a high-level research team shares progress, challenges ideas constructively, documents decisions, and contributes with excellence and collaboration.

Thank you

for the mentorship,
trust, and shared
spirit of excellence.



VIDA at DePaul • Summer 2026 research experience

SUPPLEMENTAL APPENDIX

Supporting methods, schemas, diagnostics,
and complete references

Slides 1–25 are the main presentation. Detailed methods, schemas, and supplemental results follow.

Decision log 3/3: modeling and evaluation

Selection, thresholds, uncertainty, and the secondary analysis were separated from the held-out test.

Decision	Choice used in this build	Evidence or status
Comparison design	Seven feature-block experiments × four algorithms = 28 models; M0, M1, and M2 share cohort and outcome.	Prespecified nested-comparison design
Model candidates	Logistic C=.1/1/10; HistGB 3 lr/leaves/L2 settings; RF and XGBoost each used 3 candidate settings.	Project candidate grid; random seed 20260812
Training workflow	Median imputation fitted in each pipeline; choose by validation AUROC; refit train+validation; test once.	Leakage-control protocol; no test-informed selection
Operating thresholds	Evaluate fixed .50, validation Youden, and validation thresholds targeting 80% and 90% sensitivity.	Thresholds select operating points; they do not change AUROC
Uncertainty + secondary	Patient bootstrap: 200 primary, 500 secondary. Secondary used readmitted cases, M2 index features, and original split.	Project protocol; Kruskal–Wallis then Dunn–Holm for return time

EDA established the usable fields, data grain, and feature states

Source profiling was completed before cohort construction, diagnosis mapping, SDOH encoding, and model testing.

ENCOUNTER CSV

7,675,801 rows • 31 columns

Build-used columns

EncounterKey · PatientDurableKey
Date · PrimaryDiagnosisKey
Type · VisitTypeDescription · DepartmentKey
IsEdVisit · IsHospitalAdmission
IsInpatientAdmission · IsObservation

Key audit statistics

99,439 hospital admissions → 97,310 eligible
59,449 patients • 11,973 readmissions (12.30%)

DIAGNOSIS CSV

1,531,262 rows

Columns inspected

DiagnosisKey · DiagnosisValue
GroupCode · GroupName · DiagnosisName

Uniqueness and mapping checks

1,382,556 unique DiagnosisKeys
129,173 reused/ambiguous keys flagged
99,261 distinct ICD-10 values
PrimaryDiagnosisKey → DiagnosisKey → ICD-10 → CCSR

SDOH CSV

3,977,901 response rows

Columns inspected

DisplayName · AnswerText · Domain
EncounterKey · PatientDurableKey

Coverage and raw values

305,351 patients • 1,417,888 encounters
32 question labels • 125 raw answers
Examples: Never, No, Yes, Former, Current,
Patient declined, Patient unable to answer, Unknown
12 raw Domain values, including NA*

EDA workflow before modeling

1 Schema & grain

Confirm encounter-level keys, column types, and source-table roles.

2 Completeness

Profile missingness, distinct values, duplicates, and conflicting keys.

3 Eligibility

Check Date and admission flags; require a complete 30-day follow-up window.

4 Join QA

Validate patient–encounter keys and diagnosis/SDOH linkage without row expansion.

5 Encode & verify

Preserve risk, negative, unable, nonresponse, no-record, and conflict states.

*The raw Domain field included NA; smoking-related responses were reconciled separately. Raw answers were converted to analysis states rather than treated as equivalent negatives.

Most readmissions did not repeat the index diagnosis

Separate outcome-positive analysis · 11,973 readmitted index encounters linked to the next hospital admission

Index-to-readmission relationship	n	Percent	Median days	IQR, days
Exact ICD-10 match	2,045	17.1%	13	6–21
Same CCSR category	1,315	11.0%	12	5–20
Same body system	3,078	25.7%	9	4–17
Different body system	5,128	42.8%	14	7–21
Unknown	407	3.4%	1	0–5

98.7%

had a resolvable
readmission ICD-10
diagnosis

42.8%

returned in a different body
system

Descriptive

analysis; no causal or
inferential claim

No leakage

readmission diagnosis was
never a predictor

Tree-model M1 retains all 20 body-system columns

Logistic regression excludes ccsr_system_cir so circulatory diagnoses form the reference category.

ccsr_system_bld

ccsr_system_cir

ccsr_system_dig

ccsr_system_ear

ccsr_system_end

ccsr_system_eye

ccsr_system_fac

ccsr_system_gen

ccsr_system_inf

ccsr_system_inj

ccsr_system_mal

ccsr_system_mbd

ccsr_system_mus

ccsr_system_neo

ccsr_system_nvs

ccsr_system_pnl

ccsr_system_prg

ccsr_system_rsp

ccsr_system_skn

ccsr_system_sym

Reference coding keeps interpretation stable while tree models retain the complete indicator set.

The secondary analysis remains leakage-safe

Question and cohort

- Start with the 11,973 qualifying 30-day readmissions linked by PatientDurableKey and EncounterKey.
- Exclude 407 Unknown relationships from the primary rank test and binary prediction target.
- Positive outcome: different-body-system readmission. Negative outcome: exact ICD-10, same CCSR, or same body system.
- Days-to-readmission comparisons use Kruskal–Wallis followed by Dunn pairwise tests with Holm correction.

Prediction workflow

- 1 Predictors**
Index-available M2 only: prior utilization, index principal-diagnosis EHR, and pre/index SDOH
- 2 Partitions**
Original patient-separated train, validation, and held-out test assignments
- 3 Selection**
Original three-setting candidate grids; choose by validation AUROC
- 4 Evaluation**
Refit train + validation; test once; 500 patient-bootstrap AUROC replicates

Known diagnosis relationships: n = 11,566 encounters

Illustrative M0 rows

Synthetic examples show how the single predictor aligns with outcome and split controls.

Patient	Encounter	High prior utilization	Readmitted in 30 days	Partition
P0001	E1001	1	1	train
P0002	E1002	0	0	train
P0003	E1003	1	0	validation
P0004	E1004	0	1	test

Illustrative rows only; they are not patient records.

Illustrative M1 rows

Selected columns demonstrate body-system one-hot encoding and diagnosis-quality states.

Encounter	High use	Circulatory	Digestive	Respiratory	Mental/behavioral	Dx missing	Unclassified	Outcome
E1001	1	0	0	1	0	0	0	1
E1002	0	1	0	0	0	0	0	0
E1003	1	0	1	0	0	0	0	0
E1004	0	0	0	0	0	1	1	1

Illustrative rows only. The circulatory column is absent from logistic X.

Illustrative M2 rows

Selected features demonstrate how diagnosis and SDOH states coexist in the final matrix.

Encounter	High use	Resp. Dx	Food risk	Food no record	Housing risk	Transport unable	Depression risk	Alcohol nonresponse	Outcome
E1001	1	1	1	0	0	0	1	0	1
E1002	0	0	0	0	1	0	0	1	0
E1003	1	0	0	1	0	0	0	0	0
E1004	0	0	0	0	0	1	—	—	—

Illustrative rows only. Em dashes mark values not supplied for the final E1004 example.

M2 SDOH domains – 1 of 2

Representative feature names show the domain-specific state encoding.

Domain	Example columns
Alcohol	sdoh_alcohol_risk · unable · no_record
Depression	sdoh_depression_risk · nonresponse
Financial strain	sdoh_financial_strain_risk · conflict_partial
Food insecurity	sdoh_food_insecurity_risk · unable
Housing instability	sdoh_housing_instability_risk · no_record
Intimate partner violence	sdoh_ipv_risk · nonresponse

M2 SDOH domains – 2 of 2

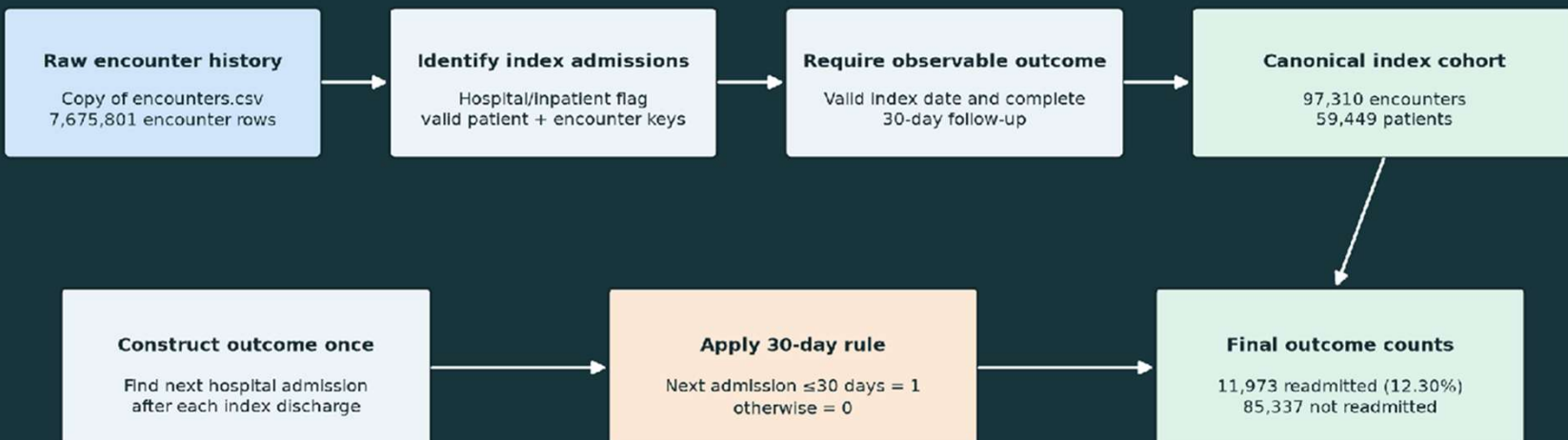
Representative feature names show the remaining domain-specific states.

Domain	Example columns
Physical activity	sdoh_physical_activity_risk · unable
Smoking	sdoh_smoking_nonresponse · no_record
Social connection	sdoh_social_connection_risk · conflict_partial
Stress	sdoh_stress_risk · unable
Transportation	sdoh_transportation_risk · no_record
Utilities	sdoh_utilities_risk · unable

Cohort construction fixes the denominator once

How the 97,310-encounter cohort was constructed

one row = one qualifying index hospitalization



Checks: 97,310 unique PatientDurableKey + EncounterKey pairs • complete follow-up for every row • no missing/invalid outcome

The same 97,310 eligible encounters and 30-day outcome flow into every downstream comparison.

M0 establishes the leakage-safe baseline matrix

M0 matrix — baseline prior utilization

97,310 rows



Predictive feature schema

Prior_High_Combined_Utilizer_365d ∈ {0,1}

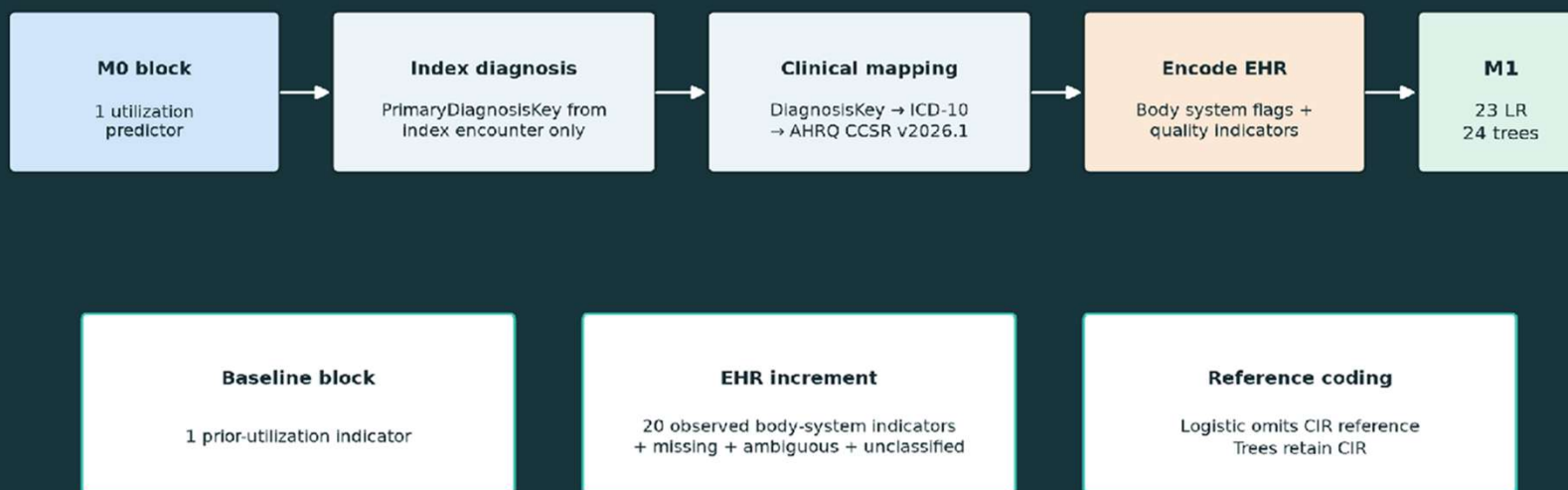
Audit-only fields such as the raw prior count and outcome dates were retained outside X and were not model predictors.

A prior-365-day combined acute-care indicator forms the common baseline before diagnosis or SDOH enrichment.

M1 adds diagnosis context through CCSR

M1 matrix — baseline + index-diagnosis EHR

97,310 rows



No readmission diagnosis was used: only the principal diagnosis known for the index encounter by discharge.

Principal ICD-10-CM diagnoses are resolved to AHRQ CCSR categories and body-system indicators.

M2 adds encounter-linked SDOH states

M2 matrix — baseline + EHR + SDOH

97,310 rows



Every M1 encounter matched exactly one encounter-level SDOH row; no post-index responses were permitted.

Twelve screening domains are reduced to explicit states and joined on patient plus encounter identifiers.

SDOH states preserve missingness and response mechanism

Available states vary by domain; categories are not collapsed into “negative.”

SDOH state	Interpretation
risk	Available response indicated the adverse condition
negative_reference	Substantive response indicated no identified risk
unable	Patient was unable to answer or complete the item
nonresponse	Screening record existed without a substantive answer
no_record	No qualifying screening information was available
conflict_partial	Records were conflicting or only partially classifiable

The full readmission model used 87 index-available features

87

M2 tree / XGBoost specification | exact feature names

01-29

01 Prior_High_Combined_Utilizer_365d
02 ccsr_system_bld
03 ccsr_system_cir
04 ccsr_system_dig
05 ccsr_system_ear
06 ccsr_system_end
07 ccsr_system_eye
08 ccsr_system_fac
09 ccsr_system_gen
10 ccsr_system_inf
11 ccsr_system_inj
12 ccsr_system_mal
13 ccsr_system_mbd
14 ccsr_system_mus
15 ccsr_system_neo
16 ccsr_system_nvs
17 ccsr_system_pnl
18 ccsr_system_prg
19 ccsr_system_rsp
20 ccsr_system_skn
21 ccsr_system_sym
22 diagnosis_missing
23 diagnosis_ambiguous
24 ccsr_body_system_unclassified
25 sdoh_alcohol_risk
26 sdoh_alcohol_negative_reference
27 sdoh_alcohol_unable
28 sdoh_alcohol_nonresponse
29 sdoh_alcohol_no_record

30-58

30 sdoh_alcohol_conflict_partial
31 sdoh_depression_risk
32 sdoh_depression_negative_reference
33 sdoh_depression_nonresponse
34 sdoh_depression_no_record
35 sdoh_financial_strain_risk
36 sdoh_financial_strain_negative_reference
37 sdoh_financial_strain_unable
38 sdoh_financial_strain_nonresponse
39 sdoh_financial_strain_no_record
40 sdoh_financial_strain_conflict_partial
41 sdoh_food_insecurity_risk
42 sdoh_food_insecurity_negative_reference
43 sdoh_food_insecurity_unable
44 sdoh_food_insecurity_nonresponse
45 sdoh_food_insecurity_no_record
46 sdoh_food_insecurity_conflict_partial
47 sdoh_housing_instability_risk
48 sdoh_housing_instability_negative_reference
49 sdoh_housing_instability_unable
50 sdoh_housing_instability_nonresponse
51 sdoh_housing_instability_no_record
52 sdoh_housing_instability_conflict_partial
53 sdoh_ipv_risk
54 sdoh_ipv_negative_reference
55 sdoh_ipv_unable
56 sdoh_ipv_nonresponse
57 sdoh_ipv_no_record
58 sdoh_ipv_conflict_partial

59-87

59 sdoh_physical_activity_risk
60 sdoh_physical_activity_negative_reference
61 sdoh_physical_activity_unable
62 sdoh_physical_activity_nonresponse
63 sdoh_physical_activity_no_record
64 sdoh_smoking_nonresponse
65 sdoh_smoking_no_record
66 sdoh_social_connection_risk
67 sdoh_social_connection_negative_reference
68 sdoh_social_connection_unable
69 sdoh_social_connection_nonresponse
70 sdoh_social_connection_no_record
71 sdoh_social_connection_conflict_partial
72 sdoh_stress_risk
73 sdoh_stress_negative_reference
74 sdoh_stress_unable
75 sdoh_stress_nonresponse
76 sdoh_stress_no_record
77 sdoh_transportation_risk
78 sdoh_transportation_negative_reference
79 sdoh_transportation_unable
80 sdoh_transportation_nonresponse
81 sdoh_transportation_no_record
82 sdoh_transportation_conflict_partial
83 sdoh_utilities_risk
84 sdoh_utilities_negative_reference
85 sdoh_utilities_unable
86 sdoh_utilities_nonresponse
87 sdoh_utilities_no_record

Feature blocks: prior utilization | index diagnosis/EHR | pre/index SDOH

Identifiers, outcome, and partition controls excluded

TIME TO READMISSION

Time to return differed across diagnosis relationships

$$H(3) = 346.54 \cdot p < 0.001$$

Relationship	n	Median	IQR
Different body system	5,128	14 days	7–21
Exact ICD-10 match	2,045	13 days	6–21
Same CCSR category	1,315	12 days	5–20
Same body system	3,078	9 days	4–17

Dunn–Holm pair	Adjusted p
Different vs exact ICD	<0.001
Different vs same CCSR	<0.001
Different vs same system	<0.001
Exact ICD vs same CCSR	0.052
Exact ICD vs same system	<0.001
Same CCSR vs same system	<0.001

Five of six adjusted pairwise comparisons were significant; exact ICD and same-CCSR return times were not distinguishable after correction.

Kruskal–Wallis and Dunn–Holm tests; Unknown excluded from primary analysis

Reference 3 a" LOINC social-determinants terminology

SDOH terminology authority

FULL CITATION

Regenstrief Institute. (2018). Represent social determinants of health with LOINC. <https://loinc.org/sdh/>

USEFULNESS TO THIS PROJECT

a Supports standardized identification of SDOH questions and answer lists, making encounter-linked screening variables traceable across EHR extracts.